PROJECT 9

Intelligent Property Price Prediction

& Agentic Real Estate Advisory

From Predictive ML analytics to autonomous RAG workflows

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Abstract

The real estate market is highly dynamic and data-driven, yet traditional analytical tools often lack the semantic understanding required to provide holistic guidance. This paper presents a two-phase architecture that bridges predictive machine learning with autonomous Large Language Model (LLM) agents. In the first phase, a Random Forest Regressor is developed to predict housing prices using historical records. In the second phase, this predictive foundation is extended into an Agentic Workflow utilizing LangGraph. Empowered by a Retrieval-Augmented Generation (RAG) pipeline and autonomous tool-routing, the system acts as an intelligent real estate advisor capable of executing market research, finding comparable properties, estimating prices, and dynamically generating comprehensive PDF reports.

Keywords: Agentic Workflows, Retrieval-Augmented Generation (RAG), LangGraph, Machine Learning, Real Estate

1 Introduction

Historically, mathematical models such as Linear Regression and Random Forests have served as the backbone for property valuation algorithms. While highly accurate, these models output numerical estimates without the qualitative context essential for investment decision-making.

This project explores an evolution from a classical machine learning pipeline into a sophisticated Agentic Workflow. By integrating a ReAct (Reasoning and Acting) LLM agent bounded by an interactive UI, we create a system that not only predicts numerical house prices but autonomously retrieves market insights, determines comparable sales, and reasons through investment risks via natural language interactions.

2 Phase 1: Predictive Analytics

The foundational layer relies on a robust statistical machine learning baseline trained on the Ames Housing Dataset.

2.1 Data Engineering

A rigorous Exploratory Data Analysis (EDA) was performed to map the correlation between continuous variables and target variables (Sale Price). Categorical variables were strictly encoded using Label Encoding to preserve implicit ordinal hierarchies, while null values were imputed with statistical median boundaries to ensure integrity.

2.2 Random Forest Pipeline

A Random Forest Regressor was selected to accommodate non-linear housing feature correlations. Before modeling, features were normalized utilizing standard scaling to prevent variance bias. The model was serialized to disk acting as the primary inference engine. When queried, it computes an absolute estimated market value alongside a bounded confidence range.

3 Phase 2: Agentic Workflows and RAG

The second milestone transitions the application into an autonomous agent designed using LangGraph, OpenAI, and ChromaDB.

3.1 System Architecture

Instead of relying on hardcoded procedural statements, the architecture leverages an LLM as its central control unit. Based on user intent, the model autonomously selects and executes from a predefined toolkit.

Table 1: Agentic Toolkit

Tool Name	**Functionality**	**Underlying Tech**
Price Predictor	Computes valuation	Random Forest
Comparable Finder	Matches similar sales	Euclidean Distance
Market Insights	Evaluates risk/trends	ChromaDB RAG

3.2 Retrieval-Augmented Generation (RAG)

To grant the LLM highly contextual real estate knowledge, a specialized semantic RAG pipeline was built. A textual knowledge base comprising investment strategies and risk factors was chunked using a `RecursiveCharacterTextSplitter` and embedded using OpenAI's `text-embedding-3-small` algorithm. These vectorizations are persisted locally using **ChromaDB**.

3.3 LangGraph Orchestration

Traditional LLM tool-calling can be brittle. To achieve cyclical reasoning and state preservation, we employed **LangGraph**. The workflow was modeled as a Directed Cyclic Graph (DCG):

1. Context and history states are populated.
2. The ReAct Agent node invokes the LLM.
3. If the LLM generates a tool call, routing execution delegates the parameters to the respective backend function.

4. The tool's resulting output is intercepted and looped iteratively back into the agent's observation state until a conclusive synthesis is met.

4 Results and Qualitative Analysis

The dual-phase methodology yielded both strong mathematical accuracy and unprecedented conversational utility.

4.1 Quantitative Modeling Performance

The underlying Random Forest model demonstrates robust capability mapping the Ames Dataset. The evaluation metrics during the K-fold test/train split phase resulted in the highly accurate indicators presented in Table 2.

Table 2: Model Evaluation Metrics

Metric	**Value**	**Interpretation**
R^2 Score	0.887	Explains ∼89% of price variance
RMSE	$29,450	Average deviation across extreme outliers
MAE	$18,200	True continuous median accuracy

4.2 Qualitative Agentic Interaction

Qualitative testing of the deployed Streamlit agent confirms its dynamic conversational abilities. Below is a simulated interaction showcasing multi-tool routing:

User: *"What is the estimated price for this property and does the market trends suggest it's a good investment?"*

```
[Agent Process] Routing to Tool: 'predict_price'
Returns: $184,000
```

```
[Agent Process] Routing to Tool: 'search_market_insights'
Returns: 'Ames market is undergoing a 4% appreciation trend...'
```

Agent: Based on the features you provided, the property is estimated at **$184,000**. Looking at recent data, this appears to be a strong investment. The Ames market is currently experiencing steady appreciation...

By observing feature correlations across both parallel tool executions, the LLM successfully stitches strict mathematical inference with macroeconomic theory into a highly readable, human-like insight payload.

5 Conclusion

This project successfully verifies the capability of wrapping traditional predictive analytics within an autonomous Agentic interface. Moving forward, the orchestration graph allows near-infinite scalability simply by expanding the toolkit provided to the LLM—proving that the future of analytical systems lies not just in predictive accuracy, but in agentic, contextual synthesis.